

Article

The Taxonomic and Functional Diversity of Leaf-Litter Dwelling Ants in the Tropical Dry Forest of the Colombian Caribbean

Jose Camargo-Vanegas , Sebastian de la Hoz-Pedraza , Hubert Sierra-Chamorro and Roberto J. Guerrero * 

Universidad del Magdalena, Santa Marta 470003, Colombia; josecamargo@unimagdalena.edu.co (J.C.-V.); sebastianhozp@unimagdalena.edu.co (S.d.l.H.-P.); hubertsieraac@unimagdalena.edu.co (H.S.-C.)

* Correspondence: rguerrero@unimagdalena.edu.co

Abstract: There have been few advances in understanding the organization and dynamics of ants in tropical dry forests. The latter are a seriously threatened ecosystem, and ants are important indicators of diversity, disturbance, and restoration in forest ecosystems. Using diversity data and morphofunctional traits, we evaluated the spatial and temporal variation of taxonomic and functional ant groups; in addition, we explored the variation in functional traits and diversity among communities. Ants were sampled during the dry and rainy seasons using mini-Winkler bags. A total of 9 subfamilies, 57 genera, and 146 species were collected. Ant species composition and richness varied both spatially (75 to 119 species) and temporally (121 and 127 species). The fragments from N2 and N3 showed higher diversity than those from N1. The dissimilarity among all areas was moderate (50–60%), mainly attributable to species turnover processes (77%). Twenty functional groups were identified. The N3 fragments had the highest functional diversity, with lower resistance to species loss, while the N1 and N2 fragments reduced functional diversity and increased similarity among species. Our results highlight the importance of integrating a functional analysis with the taxonomic assessment of ants as an important contribution to understanding the organization and dynamics of this community of insects that inhabit the tropical dry forest.



Citation: Camargo-Vanegas, J.; de la Hoz-Pedraza, S.; Sierra-Chamorro, H.; Guerrero, R.J. The Taxonomic and Functional Diversity of Leaf-Litter Dwelling Ants in the Tropical Dry Forest of the Colombian Caribbean. *Diversity* **2024**, *16*, 687. <https://doi.org/10.3390/d16110687>

Academic Editors: Merav Seifan and Luc Legal

Received: 19 August 2024

Revised: 24 October 2024

Accepted: 30 October 2024

Published: 11 November 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: functional groups; functional homogenization; spatial differentiation; species richness; species turnover

1. Introduction

The tropical dry forest (TDF) is defined as an ecosystem with low humidity and precipitation, which is distributed in the lowlands of tropical and subtropical regions [1,2]. Its most distinctive characteristic is the marked seasonality that generates a wide variety of morphological, physiological, and behavioral responses in the organisms that inhabit it, favoring the development of a high biological diversity and high levels of endemism, in addition to sustaining the interaction of processes and services that directly or indirectly sustain this ecosystem [1]. In Colombia, the TDF is one of the most threatened ecosystems, the latter resulting from the development of multiple anthropogenic activities, such as the expansion of the agricultural frontier, livestock production, and human settlement [3,4], which have increased the deterioration of the functionality and services that this terrestrial ecosystem provides [5].

The Colombian Caribbean presents the largest extension of TDF (417,838 ha) with the highest proportion in Cesar, Bolívar, and Magdalena [1,6]. In the west of this region, fragments of TDF are found in the Montes de María (MM) and the Serranía de Piojo (SP); the first ones are located between the departments of Bolívar and Sucre, while the latter are in the department of Atlántico. These fragments are surrounded by grasslands, agricultural crops, bread plantations, and forests in succession. The drastic reduction and rapid transformation of the TDF have led to a loss of biological diversity at all scales, directly affecting the ecological processes that ensure the functionality of the ecosystem

and therefore the services it provides to society, such as carbon capture, nutrient cycling, soil protection from erosion, water regulation, among others [1,7,8].

Ants are among the biotic elements of the TDF ecosystem that can suffer the negative effects of the loss of vegetation cover (canopy reduction and decreased leaf-litter production). These social insects constitute one of the most important groups of insects at an ecological level because they participate in a wide variety of processes, such as the dispersal and elimination of seeds, predation, the incorporation of nutrients into the soil system, and the establishment of vegetation [9–11]. Furthermore, the ecological qualities of ants, such as high specialization, ecological fidelity, environmental sensitivity, and high abundance [11–13], allow for the study of their communities to be very useful for the evaluation of changes in terrestrial environments [14,15], positioning them as important indicators of the diversity, disturbance, and restoration of forest ecosystems [11]. Despite this multiplicity of functions and ecological contribution to the ecosystem, there is little progress in understanding the organization and dynamics of ant communities in the TDF ecosystem. Therefore, it is necessary to integrate information from the different dimensions of diversity by adding the attributes of the species (i.e., identity, abundance) and their influence on ecosystem functions (i.e., traits, functional groups) [16].

Understanding the ecological structure and function of ant species in the TDF, as well as their responses to environmental changes, can be assessed by measuring functional traits [16–18]. These traits include phenotypic characteristics of species that determine their response to environmental changes or the effect they have on ecosystem processes [16,19], demonstrating a close relationship with biotic interactions, abiotic filters, and ecological processes [20–22]. In this sense, considering the role of ants as ecosystem engineers, their traits could have important implications in ecosystem processes [23,24]. In fact, several studies have shown strong relationships between functionally important traits in ants, habitat complexity, and disturbances [25–27], contributing to the understanding of community structure, their interactions, and processes in ecosystems.

Functional traits can be classified into hard traits and soft traits. Hard traits directly assess ecological function and are closely related to the physiology of organisms, while soft functional traits indirectly influence the ecological function of species through the measurement of certain morphological characteristics [28]. In the functional ecology of ants, morphological traits have received increased attention, as morphological variation reflects the different ways in which organisms interact with each other and with the environment, providing crucial information about the diverse ecological strategies they use to coexist under different environmental conditions [23,29,30]. This offers two main advantages: (1) morphological traits are relatively easy to measure for a significant number of species, even for those undescribed or little known [22], making them a common tool for comparison between studies on a global scale, and (2) morphological traits are associated with the ecological function performed by the species [22,30], which allows for a better understanding of the functionality of communities and ecosystems [16,23]. For example, hindlimb length in ants has been negatively correlated with leaf-litter cover on the forest floor, i.e., long hind legs make it difficult to search for and exploit food resources within the leaf litter [31]. Weber length has also been indicative of habitat complexity, such as the leaf litter, where a larger body size and relatively long legs may hinder movement [32,33], and mandibular length has been linked to diet type and habitat heterogeneity, where longer mandibles may be key to attack larger prey [22,34].

Integrating morphological traits into community categorization assumes that species with similar traits perform similar functions in their community or might be exposed to similar environmental filters [30]. However, it is important that the information underlying the traits can be linked to aspects of natural history (e.g., functional groups) to provide a complete picture of an organism's role in the ecosystem [26,30].

In the Neotropical region, mainly Brazil, studies on ants in TDF ecosystems have been conducted to evaluate how seasonal and successional effects of vegetation influence community structure [35,36]. Likewise, several studies have demonstrated a positive rela-

tionship between taxonomic and functional diversity along gradients of forest succession and land use changes [26,37]. In Colombia, studies on TDF ants are mainly concentrated in the western and southwestern regions of the Colombian territory [38–42], while few studies on taxonomic diversity in ants in the TDF of the Caribbean region have been oriented to the generation of taxonomic inventories and diversity assessments [43–46], and only a couple of cases have studied the functional structure of ant communities [47,48]. The lack of information combined with the high curatorial liability of the material (e.g., low taxonomic resolution, a scarcity of resources with associated metadata, a lack of a large reference collection, and the digitalization of the information) have not allowed for progress in the understanding of the different levels of the diversity of ants, undermining the usefulness of the data and, consequently, limiting the understanding of ecological patterns and processes at local and regional scales in the TDF.

In this work, we evaluate the spatial and temporal variation in taxonomic diversity and functional groups of ants in tropical dry forest fragments distributed in the northwestern Colombian Caribbean. In addition, we explore the variation in functional traits and diversity among ant communities that inhabit the different fragments. This information allows us to advance in the understanding of the organization and dynamics of ants in the TDF.

2. Materials and Methods

2.1. Study Area

This study was carried out at six sites with TDF fragments distributed in three areas (Figure 1): (1) TDF fragments in Atlántico into Serranía de Piojó (SP); (2) TDF fragments in Bolívar; and (3) TDF fragments in Sucre into the Montes de María (MM) and Serranía de Piojó (SP). The Montes de María subregion is composed of a mountain system with numerous bodies of water, which shape the climatic regime from dry to humid, with temperatures between 26 °C and 30 °C, average rainfall of 1500 mm, and relative humidity between 75% and 85% [49]. On the other hand, the department of Atlántico presents an enclave of low mountains that corresponds to the district of the Serranía de Piojó. The latter has a tropical climate of the steppe and savannah, with both semi-arid and semi-dry features, with average annual temperatures above 24 °C and an average annual rainfall of 1200 mm. The Montes de María subregion and the Serranía de Piojó district have extensions of TDF that belong to the biogeographic province of the Peri-Caribbean arid belt [50].

Study areas:

1. TDF fragments in Atlántico (hereafter N1): (a) Luriza Integrated Management Regional District (10.75198 N, −75.03075). This is an area declared protected for preservation, conservation, and restoration of dry forest, with an extension of 837 ha. (b) “Palmar del Titi” Integrated Management Regional District (10.65405 N, −75.18625 W). It is a remnant of dry forest with an area of 2622 ha, with a good state of conservation, gallery forests, and secondary vegetation.
2. TDF fragments in Bolívar (hereafter N2): (a) Los Colorados Flora and Fauna Sanctuary (9.925380 N, −75.18625). It is an area declared protected for conservation and considered the best-preserved fragment of tropical dry forest in the Colombian Caribbean, with an area of 1000 ha. (b) Brasilar tropical dry forest reserve (9.908608 N, −75.187768). It is a permanent plot of dry forest, which has gallery forests and secondary vegetation.
3. TDF fragments in Sucre (hereafter N3): (a) Caracolí Civil Society Nature Reserve (9.59892 N, −75.32990). It is a tropical dry forest reserve with an area of 132 ha. (b) Serranía de Coraza and Montes de María Protective Reserve (9.52169, −75.39577). This reserve presents remnants of dry forest with an area of 6730 ha.

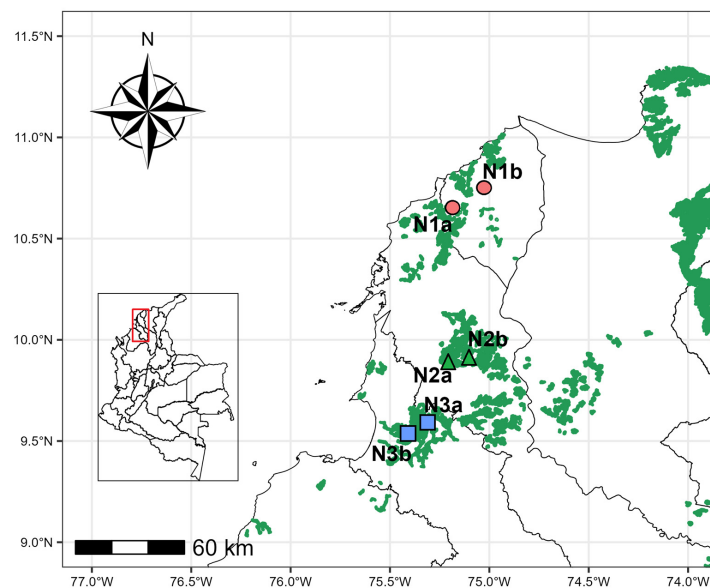


Figure 1. A map showing sampling sites of tropical dry forest in the northwest of the Colombian Caribbean. N1a: Luriza Integrated Regional Management District; N1b: “Palmar del Titi” Integrated Regional Management District; N2a: “Los Colorados” Flora and Fauna Sanctuary; N2b: Brasilar Tropical Dry Forest Reserve; N3a: “CARACOLÍ” Civil Society Nature Reserve; N3b: Coraza and Montes de María Protective Reserve. The green polygons correspond to dry forest fragments distributed both in the study area and in adjacent regions.

2.2. Ants Sampling

Two samplings were carried out in each of the six sites distributed to the northwestern Colombian Caribbean. Sampling was carried out between the period of October–December 2022 and the period of March–April 2023, covering the months of high (October–December) and low rainfall (March–April), that is, two sampling events at each site. The choice of capture method and the number of sampling units were based on the recommendations established in [51]. At each sampling site, two linear transects of 100 m were established. In each of these transects, ten 1 m² quadrants were located and separated from each other by 10 m. The collected leaf litter was sifted from each quadrant and placed in mini-Winkler bags, and each sample was placed inside independent mini-Winkler bags for 48 h. Forty samples were collected per site. Ants and other arthropods were collected in Whirl-pak[®] bags with 96% ethanol and their respective labels.

The ants and other arthropods collected in this study are covered by collection permit No. 1293 of 2013 granted to the University of Magdalena. The material collected from this work was deposited in the Biological Collections of the Universidad del Magdalena CBUMAG (RNC No. 207).

2.3. Data Analysis

Geographically close sites within their respective areas were analyzed together, considering the similarity in species richness and composition values (Figure S1; Table S1). Alpha diversity was estimated using the effective number of species from the Hill series (^qD) [52], corresponding to the integration of species richness and relative abundance to diversity measures performed by [53]. Three values of *q* were used: order 0 (⁰D, species richness), 1 (¹D, effective number of common species); and 2 (²D, effective number of dominant species) [54]. The variation of the assemblage structure was analyzed based on the shape of the range–abundance curves for ants on the litter [48]. The relative abundance of the ant species was measured as the capture frequency for each of the species, considering the arrangement of the number of litter samples in each site [48].

Estimates and comparisons of diversity were made between the different assemblages under the same or similar sampling coverage ($\hat{C}_m$). $\hat{C}_m$ corresponds to values from 0 to 1;

the closer the value is to unity, there is a high representativeness of the sampled species in relation to the total community [55]. For appropriate ecological diversity comparisons, it is important that the $\hat{C}_m$ values are similar between the areas analyzed [55,56]. For incidence-based diversity comparisons, 95% confidence intervals (95% CI) were used; non-overlapping intervals indicate significant differences [57,58]. Calculations of $\hat{C}_m$ and q-order diversity and their CIs were performed using the “iNEXT” package in R 4.4.1 software [54].

To establish spatial and temporal variation trends in the ant community, non-metric multidimensional scaling (nMDS) analysis was used based on the Bray–Curtis similarity index, using capture incidence or frequency as a measure of ant species abundance. To determine statistical differences between areas and sampling times, a multivariate analysis of variance based on permutations (Permanova) was performed, with a statistical significance level of 0.05, followed by a pairwise adonis post hoc analysis using the Bray–Curtis distance. The calculations were performed with the “betapart” package in the R software [59].

Beta diversity was estimated using the approximation of [60], whose beta diversity consists of two components: (1) species turnover (β_{jtu}) and (2) nestedness (β_{jne}). The Jaccard dissimilarity index was used to calculate the beta diversity values and its components, using the “betapart” package for R [59].

The spatial and temporal change trends of functional groups in ants were evaluated, which were identified using the classification of [27] for Neotropical ants. Additionally, other resources related to more specific aspects of the biology of the species were used [61,62].

2.3.1. Functional Traits

Eight functionally important traits in ants were selected, which have been widely related to species performance, ecological role, ecosystem processes, and interactions between various trophic levels at different scales [22,30,63,64]. The morphological traits used corresponded to standard taxonomic measures used in species descriptions, so they were relatively easy to obtain [65]. Morphological traits and their associated ecological function are provided in Table 1.

Table 1. Morphological traits used to calculate functional diversity and their ecological significance for leaf-litter-associated ant communities in study area.

Morphological Trait	Abbreviation	Functional Importance
Head length	HL	Related to the body size of the ant workers [65].
Head width	HW	Related to the size of the spaces through which ants can pass [66] and to the mandibular musculature. Wider heads have larger mandibular muscles allowing the capture of larger prey [67].
Mandible length	ML	Indicates the type of diet since longer mandibles would indicate more predatory behavior [68]; likewise, longer mandibles could allow for the capture of larger prey [34].
Eye length	EL	Related to the foraging period. It could also indicate the behavior in the search for food [38].
Interocular distance	ID	Related to hunting strategies [34] and habitat complexity [69].
Scape length	SL	Related to sensory capabilities: longer antennal scapes facilitate the tracking of pheromone trails [32].
Femur length	FL	Related to foraging speed, which reflects habitat complexity [70]. It may also be related to food quality in some specialist groups [32].
Weber length	WL	Indicative of body size, which can be related to the amount and type of resource exploited [22]. Body size can influence the microhabitats in which species forage [65]. Large-bodied ants typically forage in open conditions on the soil surface, while smaller species may occupy smaller spaces in enclosed microhabitats in leaf litter and soil [32,40].

Between one and twelve individuals per species (average = 4 individuals/species) were measured. Based on this, a greater variation within each trait was guaranteed for the most frequent species; in this way, 26 species were represented by 12 individuals, 29 species by 8 individuals, and 87 species by 4 or fewer individuals. Mean $\pm$ standard deviation (SD) of all morphological traits can be found in Table S2. Considering that morphometric traits show a strong correlation with body size, the relative measurements of each trait were determined by dividing its value by the Weber length as an approximation of the body size of each species [71]. For the analysis of the functional diversity of ant communities in the TDF fragments in each area, only the workers of each species were considered. In the case of *Pheidole* species, only the minor workers were measured. The species *Cephalotes varians* (Smith, 1876), *Camponotus* sp. 4, *Camponotus* sp. 5, and *Camponotus* sp. 6 were excluded from the analysis since only major workers were collected. Measurements were made with a dual-axis micrometer stage with output in 0.001 mm increments, but those ones were approximated to 0.01 mm due to the variable orientation of the specimens.

The community-weighted mean (CWM) of each functional trait was calculated using the average value of the trait for each species, weighted according to the respective local abundance. CWM reflects the value of the dominant trait of the community [72]. Prior to the analysis of functional diversity, all traits were transformed using the “scale” function to obtain a mean of 0 and a variance of 1, which facilitates comparisons between ant communities [64,73,74]. The distance matrix between species was calculated using the Euclidean distance, which shows a better fit for continuous or quantitative traits [75].

2.3.2. Functional Diversity

For the ant community inhabiting the TDF fragments in each area, we estimated (1) the functional richness (FRic), which measures the amount of functional space occupied by a community; (2) the functional evenness index (FEve), which expresses the homogeneity of the abundance distribution of species in a multidimensional trait space; (3) the functional redundancy index (Fred), which measures the saturation of the community’s functional space and reflects community resilience; and (4) Rao’s quadratic entropy (Rao’s Q), which incorporates the relative abundances of species as a measure of pairwise functional differences between species. Mean $\pm$ standard deviation (SD) values of all functional diversity metrics were estimated. These metrics were calculated as complementary measures of multivariate functional trait space and functional redundancy among ant communities. Community-weighted mean (CWM) values and functional trait indices among areas were compared using analysis of variance (ANOVA) followed by Tukey’s post hoc analysis. Calculations were performed with the “vegan” package in R 4.4.1 [76].

3. Results

3.1. Composition and Completeness of Sampling

A total of 30,080 individuals distributed in 9 subfamilies, 57 genera, and 146 species were collected. The subfamilies Myrmicinae, Ponerinae, and Formicinae contributed 83% of the ant richness (Figure 2). The genera with the highest number of species were *Pheidole* (19 species) and *Strumigenys* (12 species), while 24 genera were represented by a single species (Appendix A).

Solenopsis azteca Forel, 1893, was the species with the highest capture frequency (87.5%) followed by a group of six species between 50 and 76%, such as *Octostruma amrishi* (Makhan, 2007); *Pheidole flavens* Roger, 1863; and *Strumigenys eggersi* Emery, 1890. Around 79 species were in the range of 2 to 49%, within which are *Hypoponera opacior* (Forel, 1893), *Pachycondyla harpax* (Fabricius, 1804), and *Mayaponera arhuaca* (Forel, 1901). Finally, 60 species were considered rare, with frequencies below 2% (Appendix A).

Sampling coverage ranged from 0.92 to 0.96, with the lowest values occurring during the dry season. Estimates indicate that increasing the number of samples could result in collecting 11 to 41 additional species (Table S3).

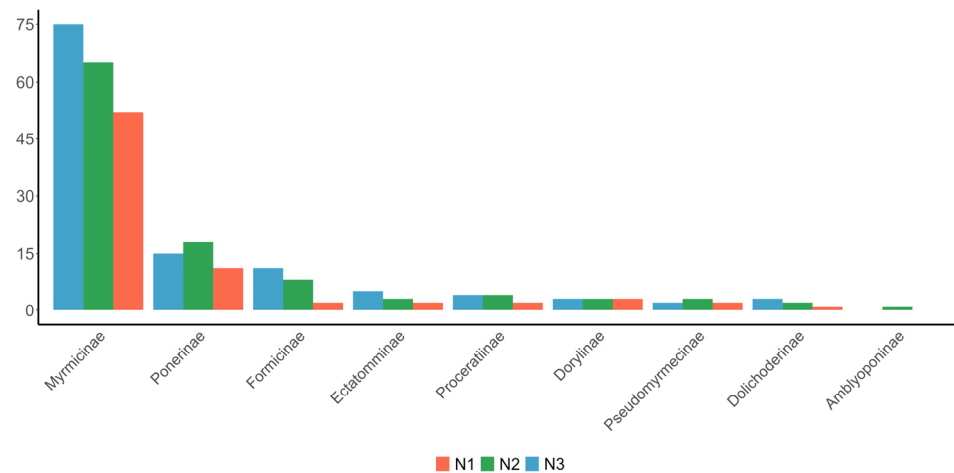


Figure 2. Number of ant species by subfamilies (horizontal axis) for TDF fragments in each area of the northwestern Colombian Caribbean.

3.2. Alpha Diversity

Species richness values are similar between both climatic periods, with 127 species recorded in the rainy season and 121 in the dry season; however, there was a notable decrease in the density of individuals collected between both seasons: 19,589 individuals were collected in the rainy season and 10,491 in the dry season.

Ant richness (0D) varied spatially and temporally. According to the 95% CI (Table S3), 0D in N1 (57 rainy season and 59 in dry season) and N2 (80 in rainy season and 85 in dry season) did not show a marked difference between both climatic periods, contrasting with N3 (103 in rainy season and 73 in dry season). The highest species diversity was found in N3 and N2; those from N1 showed the lowest in both climatic periods (Figure 3a). This trend is also observed for the 1D and 2D diversity orders in both climatic periods. In the dry season, N3 decreased their magnitudes in all three diversity orders, showing differences with respect to N1 (Figure 3).

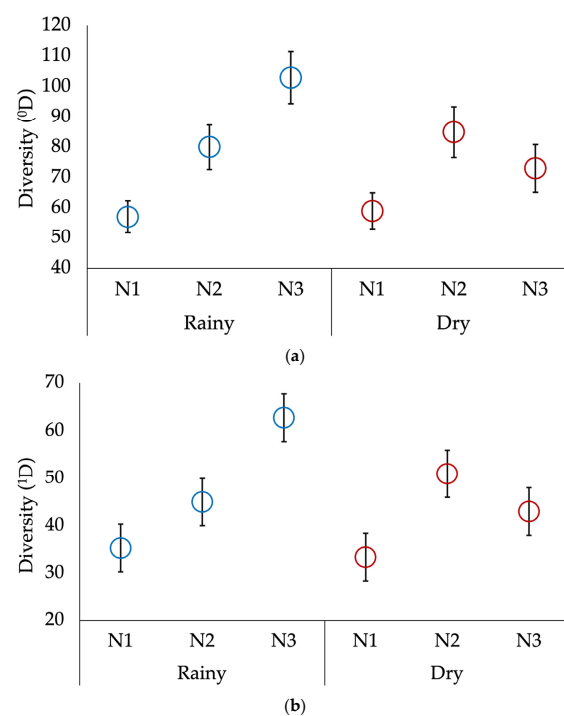


Figure 3. Cont.

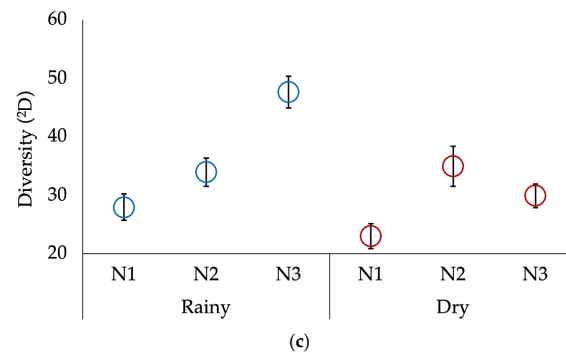


Figure 3. Diversity expressed as the effective number of ant species (2D) in TDF fragments in each area. (a) Richness (0D); (b) common species (1D); (c) dominant species (2D). The bars indicate the confidence intervals (CI) of each of the measurements.

Rank–abundance curves of ant species for the studied areas show few very frequent species (Figure 4). A total of seven species are considered frequent, which correspond to *Solenopsis azteca*, *Octostruma amrishi*, *Strumigenys eggersi*, *Pheidole flavens*, *Hypoponera opacior*, *Solenopsis geminata* (Fabricius, 1804), and *Nylanderia guatemalensis* (Forel, 1885), with capture frequencies between 50 and 82.5%. On the other hand, 79 species are rare (with capture frequencies less than 50%), and 60 are considered very rare, with capture frequencies equal to or less than 2%, such as *Alfaria minuta* Emery, 1896; *Mycocepurus curvispinosus* Mackay, 1998; *Proceratium catio* Andrade, 2003; *Gnamptogenys boliviensis* Lattke, 1995; *Acropyga fuhrmanni* (Forel, 1914); and *Rogeria curvipubens* Emery, 1894 (Appendix A).

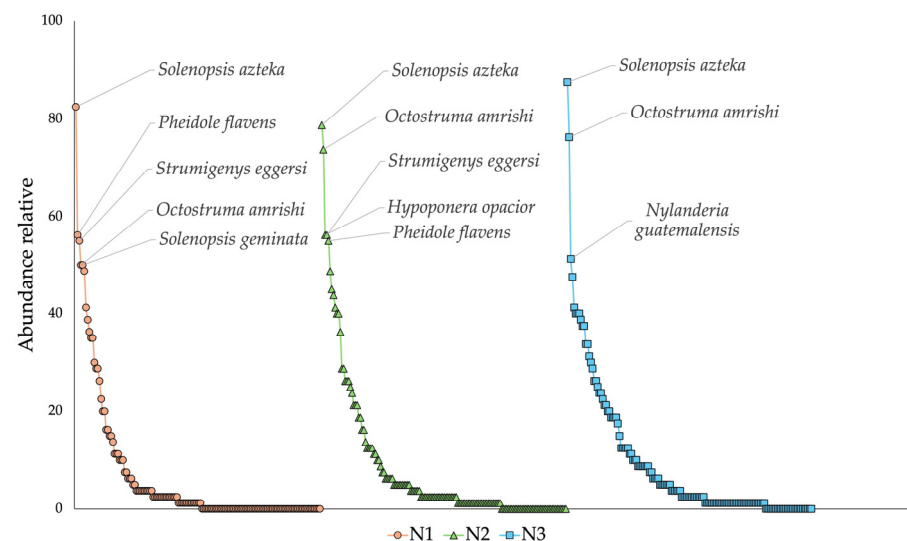


Figure 4. Rank–abundance curves showing the distribution of capture frequencies (relative abundance) for the ant assemblage in the TDF fragments in each area. The names of the species with the highest capture frequency ($\geq 50\%$) are shown.

3.3. Spatial and Temporal Variation and Beta Diversity in the Ant Community

Variations in the composition and abundance of ants living in the leaf litter in the studied areas are associated with differences between the spatial scale (Pseudo- $F_{1,9} = 2.46$; $p = 0.003$) and climatic periods (Pseudo- $F_{1,6} = 2.18$; $p = 0.0267$) (Table S4). The ordination analysis clearly shows separation in community composition at N3, whereas N1 and N2 showed overlap (Figure 5). Pairwise comparisons between areas showed significant differences between N1-N3 (Pseudo- $F_{1,6} = 3.30$; $p = 0.0034$) and N2-N3 (Pseudo- $F_{1,6} = 1.84$; $p = 0.0029$) (Table S5).

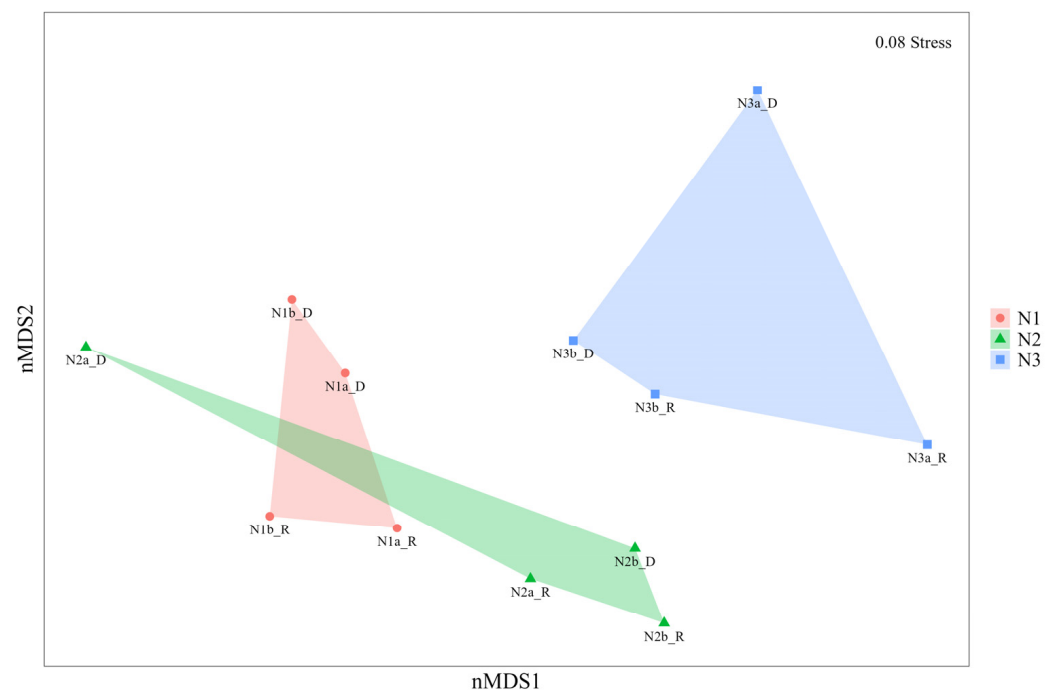


Figure 5. Ordination analysis using non-metric multidimensional scaling (nMDS) for the leaf-litter-associated ant community in the studied areas. Letters in numbers are defined in the study area section. D corresponds to the dry season, while R is the rainy season.

The analysis of beta diversity (β_{jac}) between the TDF fragments in each arc shows a moderate differentiation between 50 and 60% of dissimilarity between both climatic seasons; in general, higher dissimilarity values were found in the dry season than in the rainy season. For both seasons, turnover is the main component that explains the variation in the composition of the ants. In the dry season, turnover was 76% for N1-N2, 88% for N1-N3, and 91% for N2-N3. During the rainy season, turnover values ranged from 74 to 77% among the TDF fragments from N1-N2 and N2-N3 and 54% among N1-N3. The highest nestedness values were recorded among N1-N3 (46%) in the rainy season, while in the dry season, nestedness ranged between 9 and 24% (Figure 6) (Table S6).

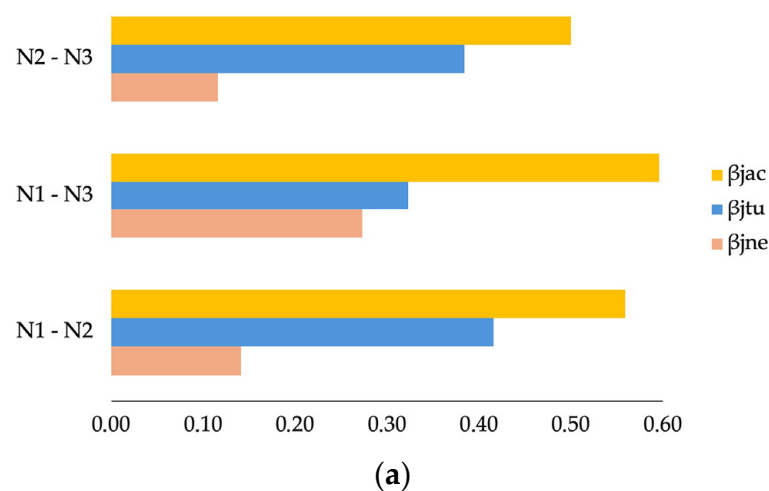


Figure 6. Cont.

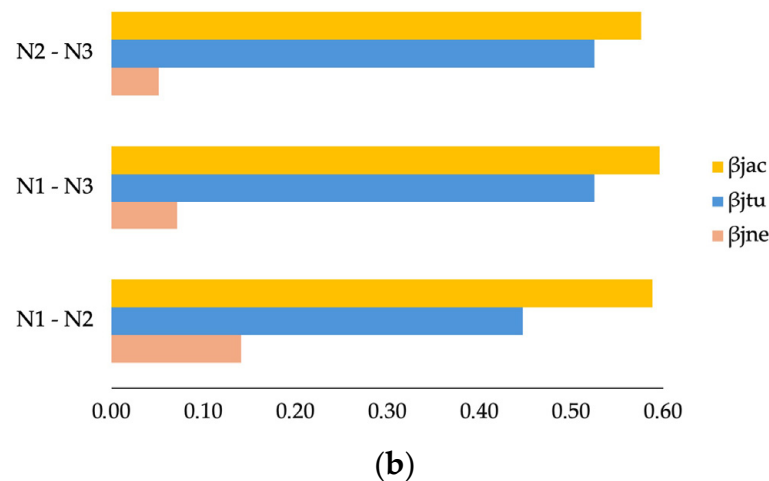


Figure 6. Partition of beta diversity (β_{jac}) into its turnover (β_{jtu}) and nestedness (β_{jne}) components between pairs of sites in each area based on the Jaccard index. (a) Rainy season; (b) dry season.

3.4. Functional Groups

Twenty functional groups were identified: eighteen were recorded in N1, nineteen in N3, and twenty in N2. The predominant functional groups in terms of species richness correspond to epigeal/litter/small hypogeal omnivores (SO), followed by arboreal omnivores (AO) and dacetine predators (DP) (Figure 7). There was no variation in the richness of functional groups between the rainy and dry seasons.

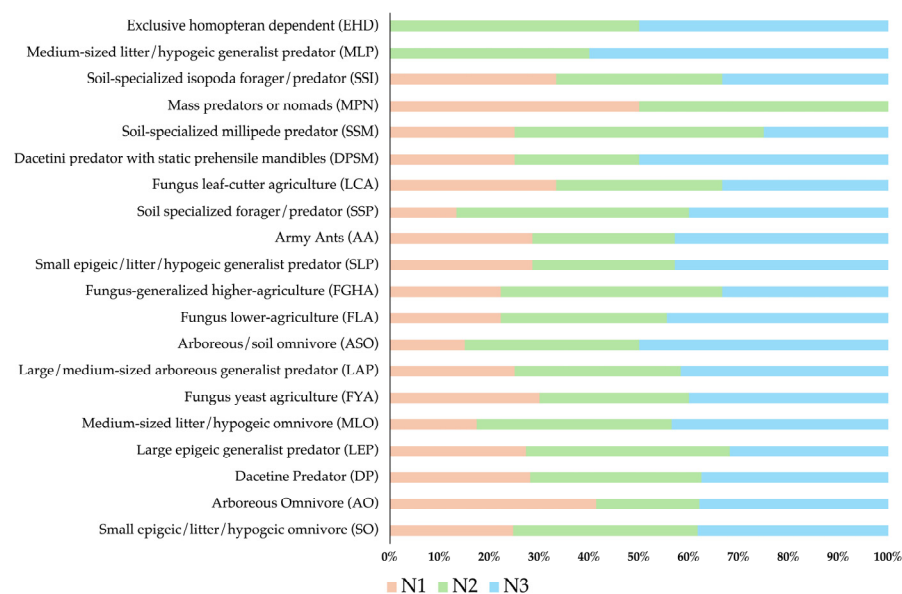


Figure 7. The spatial variation in the functional groups recorded in the TDF fragments in each area.

3.5. Functional Traits

Among the eight morphofunctional traits analyzed (Figure 8), only ML presented differences in CWM values between the studied areas (Figure 8c). The mean values of the mandible length of the ant community in N3 showed differences with those of the ants that inhabit in N2 ($0.653 \text{ mm} \pm 0.096$ and $0.536 \text{ mm} \pm 0.019$, respectively) ($F_{1,9} = 5.031$; $p = 0.047$) but not with those of the ant community in the fragments from N1 ($F_{1,9} = 5.031$; $p = 0.061$). On the other hand, the average values of the mandible length between N1 and N2 did not show significant differences ($F_{1,9} = 5.031$; $p = 0.987$) (Table S7).

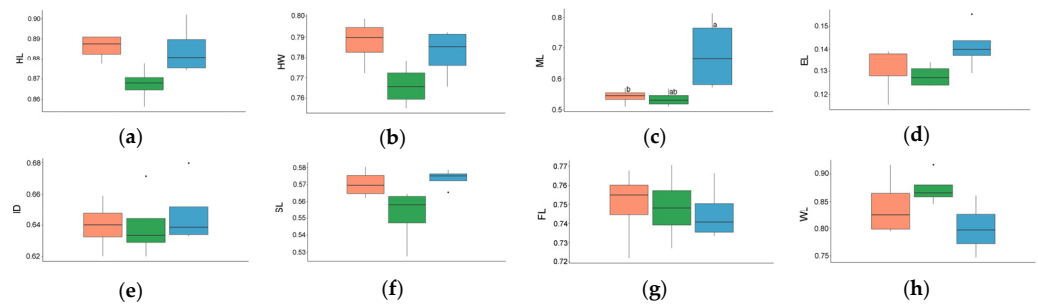


Figure 8. The functional trait weighted mean (CWM) of the eight functional traits of the ant communities located in the TDF fragments in each area: (a) head length (HL); (b) head width (HW); (c) mandible length (ML); (d) eye length (EL); (e) interocular distance (DI); (f) scape length (SL); (g) femur length (FL); and (h) Weber length (WL). In each graph, the different letters associated with the TDF fragments represent significant differences in the CWM values evaluated by Tukey's post hoc analysis. All morphometric traits are expressed in mm. The boxes represent the interquartile range between the first and third quartiles (25th and 75th percentile, respectively), and the horizontal line indicates the median. The whiskers represent the maximum and minimum values. The solid dots represent outliers. Orange boxes: N1; Green boxes: N2; Blue boxes: N3.

Functional Diversity

The functional space occupied by the ant community showed an increase in N3 (42.7 ± 21.3) by approximately five orders of magnitude relative to N2 (9.89 ± 8.22) ($F_{1,9} = 8.86$; $p = 0.018$) and seven orders of magnitude relative to N1 (5.86 ± 5.68) ($F_{1,9} = 8.86$; $p = 0.010$) (Figure 9a). Functional equitability values did not show significant differences among the TDF fragments in each area (0.65 ± 0.0436 for N3; 0.652 ± 0.036 for N2; and 0.618 ± 0.048 for N1) ($F_{1,9} = 0.76$; $p = 0.493$) (Figure 9b). N3 showed a significant reduction in functional redundancy values (0.757 ± 0.0136) compared to those from N1 (0.781 ± 0.004) ($F_{1,9} = 7.87$; $p = 0.010$) and N2 (0.775 ± 0.005) ($F_{1,9} = 7.87$; $p = 0.430$), while N1 and N2 showed no differences ($F_{1,9} = 7.87$; $p = 0.646$) (Figure 9c). Finally, N3 showed a significant increase in Rao's Q values (6.21 ± 0.92) compared to those from N2 (4.9 ± 0.22) ($F_{1,9} = 9.64$; $p = 0.021$) and N1 (4.58 ± 0.194) ($F_{1,9} = 9.64$; $p = 0.006$); on the other hand, the fragments from N1 and N2 showed no differences ($F_{1,9} = 7.87$; $p = 0.713$) (Figure 9d) (Table S8).

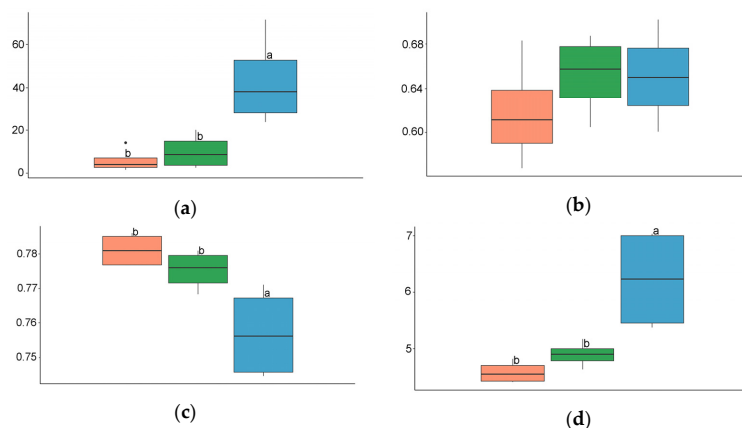


Figure 9. Estimated functional diversity indices for ant communities located in the TDF fragments in each area. (a) Functional richness (FRic); (b) functional evenness (FEve); (c) functional redundancy (Fred); and (d) Rao's quadratic entropy (Rao's Q). In each graph, the different letters associated with the TDF fragments represent significant differences in the values of each index evaluated by Tukey's post hoc analysis. The boxes represent the interquartile range between the first and third quartiles (25th and 75th percentile, respectively), and the horizontal line indicates the median. The whiskers represent the maximum and minimum values. The solid dots represent outliers. Orange boxes: N1; Green boxes: N2; Blue boxes: N3.

4. Discussion

4.1. Sampling Composition and Coverage

The ant fauna collected among TDF fragments in each area represents around 75% of the subfamilies and 42% of the genera recorded for the Neotropical region [77]; in the case of Colombia, the ant fauna reported here corresponds to 82% of the subfamilies and 54% of the genera reported [78]. Of the 146 species reported here, 123 were identified at the species level, corresponding to more than 10% of the known records for the country [79,80]. Likewise, the richness of ants that inhabit the leaf litter of the fragments studied is one of the highest records for the TDF at a regional [48,51] and national scale [41,42]. Based on the distribution of the species, 57 species are registered for the first time in N1, 58 species in N2, and 97 species in N3. On the other hand, six species are recorded for the first time in Colombia: *Gnamptogenys boliviensis* and *Megalomyrmex longinoi* Boudinot et al., 2013, living in the fragments of N2 and N3; *Apterostigma pariense* Lattke, 1997; *Cephalotes varians*; *Rogeria ciliosa* Kugler, 1994; and *Rasopone pluviselva* Longino and Branstetter, 2020, living in N3 (Appendix A).

The subfamily Myrmicinae presented the highest richness of genera and species; this is a general trend for Neotropical ecosystems due to the wide diversity of nesting habits, food resources and lifestyles shown by the species of this subfamily, including generalist predators, specialists, scavengers, omnivores, granivores, and herbivores [78,81]. This same pattern has been seen in other regions with TDF in Colombia [44–46]. The high species richness in Myrmicinae is mainly due to genera characteristic of leaf litter, such as *Strumigenys* and *Octostruma*, as well as groups that explore various habitat resources, such as the tree stratum (*Cephalotes* and *Crematogaster*) and the epigeal stratum (*Pheidole*, *Solenopsis*, and *Rogeria*). Other subfamilies in order of importance are Ponerinae, Formicinae, Ectatomminae, Proceratiinae, Dorylinae, Pseudomyrmicinae, Dolichoderinae, and Amblyoponinae; however, the contribution to the richness of these subfamilies varies among different studies [48].

Sampling coverage values interpreted in terms of completeness were high in all studied fragments, suggesting a representative sampling of the ant fauna in both climatic seasons. Sampling coverage values were lower during the dry season, indicating a greater probability that by increasing the sample size, a new sampled individual corresponds to a different species [56]. In this case, leaf-litter desiccation during the dry season could be affecting both the quality of leaf litter in the TDF, limiting the establishment of ants with a subsequent increase in the number of unique species [48].

4.2. Alpha Diversity

In this study, diversity expressed as the effective number of species (0D) showed a variable pattern at spatial and temporal levels. N2 and N3 recorded higher values of species richness compared to N1. This trend is consistent for common species (1D) and dominant species (2D) in both climatic seasons. However, variations in richness between climatic seasons were only marked in N3. The results of this study contrast with other assessments of ant diversity in the Colombian Caribbean. In the departments of La Guajira and Magdalena, a high variation in ant richness has been recorded at a spatial level, without being influenced by climatic seasons [40,43], while in the department of N1, the variation in richness is mostly explained by climatic seasons [41]. The variation in species richness between the TDF fragments among different areas could be associated with the heterogeneity and vegetation cover of the evaluated fragments, which consequently determine the contribution of leaf litter to the system [44,82]. Some results show that species-rich ant communities are associated with heterogeneous forests (e.g., developed vegetation cover, abundant leaf litter) due to a greater availability of food resources and nesting sites for the coexistence of a greater number of ant species [83–86].

The structure of the ant community in the TDF fragments in each area is represented by a low proportion of dominant species and a high proportion of uncommon and rare species. *Solenopsis azteca* was the most numerically dominant species in all studied areas. The

dominance of *Solenopsis azteca* could be explained by the nesting and foraging preferences of this species, as its generalist and opportunistic behavior allows it to take advantage of a greater variety of food resources in the leaf litter [60,87]. Another numerically dominant species is *Octostruma amrishi*, which recorded higher capture frequency values in N2 and N3 compared to the fragments of N1. Ants of the *Octostruma* genus are part of the cryptic myrmicines, so they are not usually recorded in high abundances in studies of ant diversity [88]. The high dominance of *Octostruma amrishi* is likely associated with the presence of habitats with high vegetation cover, with the presence of a deeper layer of leaf litter, which may favor the settlement of specialized species [89] as well as a wide variety of small arthropods on which they feed [27]. The other abundant species include *Strumigenys eggersi*, *Pheidole flavens*, *Hypoponera opacior*, *Solenopsis geminata*, and *Nylanderia guatemalensis*.

4.3. Spatial and Temporal Variation and Beta Diversity in the Ant Community

The composition of the ant fauna is different in space and time between TDF fragments. These differences are associated with the ant community of each area, its relative abundance (capture frequency), and the number of non-shared species. N3 has a higher proportion of exclusive species, which could be associated with the presence of microhabitats with a greater plant structure and density, promoting a greater accumulation of leaf litter [90]. Therefore, N3 would be offering a greater resource of habitability to sustain a community richer in species and with different requirements [91]. On the other hand, N1 and N2 showed no differences in ant fauna composition, possibly due to similarities in vegetation structure, leaf-litter characteristics on the forest floor, and environmental conditions between those TDF fragments [46,92]. Likewise, the high dispersal capacity of ants and the presence of living fences throughout the area could promote the dispersion of populations of different species between geographically close fragments [46,93].

Climatic periods do not show differences between species richness in N1 and N2 but do so in N3. Variation in species richness in N3 is a factor that may be influencing the observed differences in the community between climatic periods. On the other hand, humidity in the environment seems to be a limiting factor for the ant community since the number of individuals is influenced by rainfall [48]. Although we did not record humidity data at each sampling site, field observations and counting of individuals allowed us to identify some trends. During the rainy season, there was an increase in humidity in the leaf litter, promoting decomposition processes and consequently greater productivity in terms of available resources (prey, food). These characteristics have been shown to favor the establishment and growth of ant colonies [93,94]. In contrast, during the dry season a substantial decrease in plant cover was observed, resulting in the drying of the leaf litter and the slowing of the decomposition of organic matter [1,95].

High levels of beta diversity are a feature maintained in the TDF even at small geographic scales, which is probably due to environmental selection pressures associated with marked seasonality [96]. Although species turnover explains the overall pattern of variation, nestedness was substantially higher between N1 and N3 in the rainy season. N1 recorded 57 species, compared to 103 species in N3 during the rainy season (Table S3); 10 species were shared between these areas, within which are found *Acromyrmex santschii* (Forel, 1912); *Ectatomma ruidum* (Roger, 1860); *Pogonomyrmex mayri* Forel, 1899; *Strumigenys marginiventris* Santschi, 1931; and *Leptogenys ritae* Forel, 1899. Therefore, the greater contribution to nesting between N1 and N3 is explained by the differences in species richness recorded between both areas [36].

A total of 52 species are shared among the studied areas, which may be related to the similarity of ant communities within the same type of landscape [46,97]. Furthermore, one-third of the total identified species correspond to unique or exclusive species distributed throughout the study area, which underlines the importance of research at local and regional scales (Figure S2). This information constitutes an important input for the understanding of ecological patterns and processes at local and regional scales in the TDF.

4.4. Functional Groups

The functional groups with the highest species richness were consistent throughout the studied areas, highlighting the small-sized epigeal omnivores (SO) with species of the genera *Pheidole*, *Solenopsis*, *Rogeria*, *Carebara*, *Pogonomyrmex*, and *Wasmannia*. Species belonging to this group are generalists and opportunists, allowing them to take advantage of a greater variety of food resources and adapt to less favorable environmental conditions [27]. The groups of arboreal omnivores (AO) and dacetine predators (DP) were also important in terms of species richness among the studied areas; the arboreal omnivores (AO) included species from the genera *Crematogaster*, *Cephalotes*, *Dolichoderus*, *Monomorium*, *Nesomyrmex*, and *Azteca*. These species have generalist habits, whereas the majority of the species occupy the arboreal stratum, being able to nest in dead branches and actively forage in the leaf litter [27]. Dacetine predatory ants (DP) were represented by species of the genera *Strumigenys*, *Rhopalothrix*, and *Eurhopalothrix*, genera exclusive to leaf litter that feed on a wide variety of small arthropods [92]. The results highlight the importance of leaf litter as a microhabitat that offers a high availability of food resources and nesting sites that determine the taxonomic and functional structure of ant communities [90,91,98,99].

The set of functional groups in all studied areas reflect a great variability in the resources available for the establishment of numerous species of ants with different trophic requirements (herbivores, omnivores, predators, specialists). The results of this study show a homogeneous distribution of functional groups throughout the study area, where a total of 17 functional groups were shared between the TDF fragments, suggesting that despite the differences in the taxonomic composition of the ant communities, the functions performed by the species in a fragment can be assumed by other ecologically similar species in another one [48].

4.5. Functional Traits

The results suggest that mean mandible length values were higher in ants from N3 compared to ants living in N2. (Figure 8c). This trait is related to diet type with predatory ant species [34,68]: for example, Dacetini predators (DP) with mandibular specializations highly recognizable in typical leaf-litter ants [27]; specialized ground millipede predators (SSM) with elongated mandibles specialized for capturing specific prey [26]; and generalist predators of the arboreal (LAP), epigeic (LEP), and hypogeic (MLP) strata, where longer mandibles allow for predation on larger prey [34]. The results suggest an increase in the richness and abundance of long-mandible predatory ants of the genera *Strumigenys*, *Thaumatomyrmex*, *Alfaria*, *Gnamptogenys*, *Neoponera*, *Holcoponera*, and *Rasopone* in N3; this trend is maintained even in groups of predators with smaller mandibles, such as *Octostruma*, belonging to the group of Dacetini predators with static prehensile mandibles (DPSM), and other Dacetini predators (DP) of the genus *Eurhopalothrix*. The greater representation of predatory ants in N3 could be due to the presence of a deep layer of leaf litter, providing greater heterogeneity in the habitability resource of the species [90,91]. In this sense, several studies have characterized predatory ants as a group of species typical of areas with tree cover [40,100,101], being frequently associated with forest habitats where there is a greater availability of food and nesting sites as well as less competitiveness [102–104], like those found in N3. In contrast, ants from N2 recorded lower mean mandible length values, suggesting a lower contribution of large predatory ants to community structures. The results show a high proportion of small ants with hypogeic habits inhabiting N2, with specialized ground predators/foragers (SSP) of the genera *Discothyrea* and *Prionopelta* and small-sized generalist litter/hypogeic predators (SLP) of the genus *Hypoponera*.

Functional Diversity

The components of functional diversity, richness (Fric), and Rao's quadratic entropy (Rao's Q) showed a similar trend to that of taxonomic diversity, with higher values in N3. The high values of functional richness in N3 indicate a greater diversity of functional traits and, consequently, an increase in the capacity of the ant community to develop

more efficient functions in the leaf litter derived from a greater partition of the available resources and the occupation of a larger volume in the functional space [105,106]. The latter is supported by the high Rao's Q values, suggesting greater functional differences between species in these fragments. The high Rao's Q values in N3 could be related to a higher proportion of rare species (36 species) compared to N1 and N2 (14 and 26 species, respectively), indicating a more diverse community with functionally different species [102,107].

We found that N3 had a large amount of morphofunctional space ($FRic = 42.7$), but the functional redundancy decreased, suggesting that the ant community in these fragments could be less resistant to species loss [64,106–108]. A higher proportion of unique species has been directly linked to lower community resilience [105]. In this sense, species such as *Alfaria minuta*; *Eurhopalothrix pilulifera* Brown and Kempf, 1960; *Holcaponera strigata* (Norton, 1868); *Hylomyrma columbica* (Forel, 1912); *Lachnomyrmex scrobiculatus* Wheeler, 1910; *Proceratium catio*; and *Rasopone pluviselva* are shown as functionally different units from their respective traits, presenting different mechanisms for the use of resources [19,108]. The loss of these species could mean the reduction of functions intrinsically associated with the habitability resource of ants [27]. In contrast, ant communities in N1 and N2 occupy similar functional spaces ($FRic$) with a greater similarity between species (Rao's Q) favoring functional homogenization processes as well as a lower vulnerability of the community (i.e., higher values of functional redundancy, $Fred$). These results are like the pattern observed in the analysis of spatial variation, suggesting that the similarity in the taxonomic composition of the species is also supported by the presence of shared functional traits between both fragments, indicating greater resistance.

5. Conclusions

The taxonomic diversity of leaf-litter-associated ants in the studied areas show a marked spatial variation between N3 and the rest of the areas studied. At the temporal level, the differences were only marked in N3; this is also reflected in the level of species richness between climatic periods. Climate variation also influences the number of ants that exploit leaf litter, as ant abundance dropped by half during the dry season. High levels of species turnover were also recorded between the studied areas, indicating the presence of relatively different ant communities among them. The results show a homogeneous distribution of functional groups throughout the study area, suggesting that differences in taxonomic composition arise from ecologically equivalent species. N3 had greater functional diversity with less resistance to species loss, while N1 and N2 reduced functional diversity and increased similarity between species, leading to functional homogenization processes and lower vulnerability.

The patterns of taxonomic alpha diversity and functional diversity as well as beta diversity and functional groups underline the importance of integrating the functional analysis of ant communities with the taxonomic assessment of species since each component reveals different aspects within the ecosystem. The results of this study contribute to a better understanding of the organization and dynamics of ant communities as well as ecological patterns and processes at local and regional scales in the TDF.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/d16110687/s1>, This section provides information about the effective number of ant species (Figure S1), species richness (Figure S2) and values of beta diversity components (Table S1) in the tropical dry forest sampling sites. Also included are the values of the functional traits analysed in the species (Table S2) and summaries of several statistical tests applied to the samples (Tables S3–S8).

Author Contributions: R.J.G. and J.C.-V. contributed equally to the conceptualization and development of this study. The sampling design and field work were carried out by J.C.-V., R.J.G., H.S.-C. and S.d.l.H.-P. The taxonomic identification of the ants was carried out by J.C.-V., H.S.-C. and S.d.l.H.-P., while the validation was carried out by R.J.G.; data curation and analysis of the information were carried out by J.C.-V., S.d.l.H.-P. and H.S.-C. J.C.-V. prepared the first draft of the manuscript. R.J.G., H.S.-C. and S.d.l.H.-P. reviewed and edited the final version of the manuscript. R.J.G. oversaw obtaining funds for this study. All authors have read and agreed to the published version of the manuscript.

Funding: This study was funded by Ministerio de Ciencia, Tecnología e Innovación de Colombia (Minciencias), Universidad del Atlántico, and Universidad del Magdalena through contingent recovery financing contract No. 80740-157-2021 (Programas Conectando Conocimiento 2019). The APC was funded by Minciencias and Universidad del Magdalena.

Institutional Review Board Statement: Not applicable.

Data Availability Statement: The data presented in this study are openly available in [repositoryname e.g., SIB Colombia] at [<https://doi.org/10.15472/gyakqs>] (accessed on 2 August 2024).

Acknowledgments: We thank the people and institutions that allowed us to access the TDF within their farms, Juan Manuel de la Hoz (Parcela Brasilar); National Parks in the Los Colorados Flora and Fauna Sanctuary; Guadel Muñoz (Palmar del Titi); Gabriel Utria (Vereda Luriza); Cesar Álvarez and Alberto Leones (Caracolí) and Rafa Romero (La Reserva Protectora Coraza); and Grupo de Investigación de Insectos Neotropicales and Centro de Colecciones Científicas—Universidad del Magdalena, for all the logistical support. Professors José Vergara, Lina María Ramos, César Tamaris, and Bladimir Zúñiga provided advice for the development of the ecological analyses carried out.

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Appendix A

Table A1. List of species collected in the tropical dry forest fragments. The number indicates the frequency of captures in percentage (%). The asterisk symbol indicates species with new records for each department (*) and new records for the country (**). The functional groups correspond to small epigeic/litter/hypogeic omnivore (SO); arboreous omnivore (AO); dacetine predator (DP); large epigeic generalist predator (LEP); medium-sized litter/hypogeic omnivore (MLO); fungus yeast agriculture (FYA); large/medium-sized arboreous generalist predator (LAP); fungus lower-agriculture (FLA); fungus-generalized higher-agriculture (FGHA); small epigeic/litter/hypogeic generalist predator (SLP); army ants (AA); soil-specialized forager/predator (SSP); fungus leaf-cutter agriculture (LCA); Dacetini predator with static prehensile mandibles (DPSM); soil-specialized millipede predator (SSM); mass predators or nomads (MPN); soil-specialized isopoda forager/predator (SSI); arboreous/soil omnivore (ASO); medium-sized litter/hypogeic generalist predator (MLP); and exclusive homopteran dependent (EHD).

Taxa	Functional Groups	N1	N2	N3
Amblyoponinae				
<i>Prionopelta antillana</i> (Forel, 1909)	SSP		1.25 *	
Dolichoderinae				
<i>Azteca</i> sp. 1				1.25
<i>Dolichoderus bispinosus</i> (Olivier, 1792)	AO			2.5
<i>Dolichoderus</i> sp. 2	AO		1.25	
<i>Tapinoma ramulorum</i> Emery, 1896	SO	1.25 *	6.25 *	3.75
Dorylinae				
<i>Labidus coecus</i> (Latreille, 1802)	AA	6.25	1.25	1.25 *
<i>Labidus praedator</i> (Smith, 1858)	AA	2.5 *	2.5	5 *
<i>Nomamyrmex esenbeckii</i>	AA			1.25 *
<i>Syscia</i> sp. 1	MPN	3.75 *	2.5 *	

Table A1. Cont.

Taxa	Functional Groups	N1	N2	N3
Ectatomminae				
<i>Alfaria minuta</i> Emery, 1896	SLP			5 *
<i>Ectatomma ruidum</i> (Roger, 1860)	LEP	13.75	2.5	1.25
<i>Ectatomma tuberculatum</i> (Olivier, 1792)	LAP	1.25		1.25 *
<i>Gnamptogenys boliviensis</i> Lattke, 1995	SLP		2.5 **	3.75 **
<i>Gnamptogenys</i> sp. 2	SLP		1.25	
<i>Holcaponera strigata</i> (Norton, 1868)	SLP			8.75 *
Forminicinae				
<i>Acropyga fuhrmanni</i> (Forel, 1914)	EHD		3.75 *	5 *
<i>Acropyga smithii</i> Forel, 1893	EHD		1.25 *	1.25 *
<i>Brachymyrmex aphidicola</i> Forel, 1909	ASO		5 *	1.25 *
<i>Brachymyrmex minutus</i> Forel, 1893	ASO			7.5 *
<i>Camponotus</i> sp. 1	ASO		2.5	1.25
<i>Camponotus</i> sp. 3	ASO		5	5
<i>Camponotus</i> sp. 4	ASO	2.5		2.5
<i>Camponotus</i> sp. 5	ASO		1.25	1.25
<i>Camponotus</i> sp. 6	ASO			1.25
<i>Camponotus</i> sp. 7	ASO			1.25
<i>Camponotus striatus</i> (Smith, 1862)	ASO		2.5 *	
<i>Nylanderia guatemalensis</i> (Forel, 1885)	MLO	30 *	40 *	51.25 *
Myrmicinae				
<i>Acromyrmex santschii</i> (Forel, 1912)	LCA	10		1.25
<i>Adelomyrmex myops</i> (Wheeler, 1910)	SSP		2.5	8.75 *
<i>Apterostigma dentigerum</i> Wheeler, 1925	FLA			1.25 *
<i>Apterostigma parietense</i> Lattke, 1997	FLA			1.25 **
<i>Apterostigma pilosum</i> Mayr, 1865	FLA	3.75 *	16.25 *	6.25 *
<i>Atta colombica</i> Guérin-Méneville, 1844	LCA		3.75	
<i>Carebara audita</i> Fernández, 2004	SO		2.5 *	
<i>Carebara brevopilosa</i> Fernández, 2004	SO		10	12.5 *
<i>Carebara globularia</i> Fernández, 2004	SO	11.25 *	3.75 *	23.75 *
<i>Carebara striata</i> (Xu, 2003)	SO		8.75 *	11.25 *
<i>Carebara urichi</i> (Wheeler, 1922)	SO		2.5 *	
<i>Cephalotes atratus</i> (Linnaeus, 1758)	AO	1.25		
<i>Cephalotes columbicus</i> (Forel, 1912)	AO			1.25 *
<i>Cephalotes complanatus</i> (Guérin-Méneville, 1844)	AO	2.5 *		
<i>Cephalotes porrasi</i> (Wheeler, 1942)	AO	1.25 *		
<i>Cephalotes varians</i> (Smith, 1876)	AO			1.25 *
<i>Crematogaster brasiliensis</i> Mayr, 1878	AO	6.25 *	5 *	1.25 *
<i>Crematogaster carinata</i> Mayr, 1862	AO			1.25 *
<i>Crematogaster erecta</i> Mayr, 1866	AO	2.5 *	2.5 *	
<i>Crematogaster flavosensitiva</i> Longino, 2003	AO	2.5 *		31.25 *
<i>Crematogaster limata</i> Smith, 1858	AO	20 *	23.75 *	11.25 *
<i>Crematogaster nigropilosa</i> Mayr, 1870	AO	2.5 *	1.25 *	28.75 *
<i>Crematogaster obscurata</i> Emery, 1895	AO	3.75 *		3.75 *
<i>Crematogaster stollii</i> Forel, 1885	AO	1.25 *		
<i>Cyphomyrmex costatus</i> Mann, 1922	FYA	3.75 *	7.5 *	8.75 *
<i>Cyphomyrmex minutus</i> Mayr, 1862	FYA	35 *	41.25 *	47.5 *
<i>Cyphomyrmex rimosus</i> (Spinola, 1851)	FYA	16.25 *	5	2.5 *
<i>Cyphomyrmex</i> sp. 4	FYA			1.25
<i>Eurhopalothrix pilulifera</i> Brown and Kempf, 1960	DP			2.5 *
<i>Hylomyrma columbica</i> (Forel, 1912)	MLO			18.75 *
<i>Lachnomyrmex scrobiculatus</i> Wheeler, 1910	MLO			2.5 *
<i>Megalomyrmex drifti</i> Kempf, 1961	MLO		18.75	2.5 *
<i>Megalomyrmex incisus</i> Smith, 1947	MLO		2.5	5 *
<i>Megalomyrmex longinoi</i> Boudinot et al., 2013	MLO		2.5 **	1.25 **
<i>Megalomyrmex silvestrii</i> Wheeler, 1909	MLO	10 *	12.5	1.25 *
<i>Monomorium floricola</i> (Jerdon, 1851)	AO	2.5 *	6.25 *	30 *

Table A1. Cont.

Taxa	Functional Groups	N1	N2	N3
<i>Mycetomoellerius</i> sp. 1	FLA		1.25	
<i>Mycocepurus curvispinosus</i> Mackay, 1998	FLA		10 *	
<i>Myrmicocrypta</i> sp. 2	FLA			1.25 *
<i>Myrmicocrypta urichi</i> Weber, 1937	FLA	7.5 *	5 *	
<i>Nesomyrmex</i> sp. 1	AO	1.25 *		
<i>Octostruma amrishi</i> (Makhan, 2007)	DPSM	50 *	73.75 *	76.25 *
<i>Octostruma iheringi</i> (Emery, 1888)	DPSM			8.75 *
<i>Paratrachymyrmex bugnioni</i> (Forel, 1912)	FGHA	3.75	2.5 *	8.75 *
<i>Paratrachymyrmex cornetzi</i> (Forel, 1912)	FGHA		2.5 *	1.25 *
<i>Paratrachymyrmex irmgardae</i> (Forel, 1912)	FGHA	11.25 *		
<i>Pheidole biconstricta</i> Mayr, 1870	SO		2.5 *	2.5 *
<i>Pheidole fimbriata</i> Roger, 1863	SO		1.25 *	1.25 *
<i>Pheidole flavens</i> Roger, 1863	SO	56.25 *	55 *	40 *
<i>Pheidole mendicula</i> Wheeler, 1925	SO	38.75 *	48.75 *	25 *
<i>Pheidole pugnax</i> Dalla Torre, 1892	SO	48.75 *	45 *	26.25 *
<i>Pheidole subarmata</i> Mayr, 1884	SO	7.5 *	12.5 *	23.75 *
<i>Pheidole susannae</i> Forel, 1886	SO	2.5 *	2.5 *	1.25 *
<i>Pheidole synarmata</i> Wilson, 2003	SO	2.5 *	21.25 *	10 *
<i>Pheidole transversostriata</i> Mayr, 1887	SO	2.5 *	1.25 *	1.25 *
<i>Pheidole zeteki</i> Smith, 1947	SO	16.25 *	5 *	37.5 *
<i>Pheidole</i> sp. 1	SO	3.75	26.25	12.5 *
<i>Pheidole</i> sp. 2	SO		12.5	8.75 *
<i>Pheidole</i> sp. 4	SO	3.75	1.25	12.5 *
<i>Pheidole</i> sp. 5	SO		5	2.5 *
<i>Pheidole</i> sp. 9	SO		2.5	2.5 *
<i>Pheidole</i> sp. 12	SO		5	2.5 *
<i>Pheidole</i> sp. 13	SO		16.25	20 *
<i>Pheidole</i> sp. 17	SO	1.25		1.25 *
<i>Pheidole</i> sp. 18	SO			20 *
<i>Pogonomyrmex mayri</i> Forel, 1899	SO	10	1.25	6.25
<i>Rhopalothrix isthmica</i> (Weber, 1941)	DP		1.25 *	
<i>Rogeria belti</i> Mann, 1922	SO	1.25 *	12.5 *	12.5 *
<i>Rogeria ciliosa</i> Kugler, 1994	SO			1.25 **
<i>Rogeria curvipubens</i> Emery, 1894	SO	5 *	11.25 *	15 *
<i>Rogeria foreli</i> Emery, 1894	SO	36.25 *	28.75 *	2.5 *
<i>Sericomyrmex amabilis</i> Wheeler, 1925	FGHA		5 *	5 *
<i>Solenopsis azteca</i> Forel, 1893	SO	82.5 *	78.75 *	87.5 *
<i>Solenopsis geminata</i> (Fabricius, 1804)	SO	50	18.75	21.25 *
<i>Solenopsis picea</i> Emery, 1896	SO	20 *	26.25 *	41.25 *
<i>Stegomyrmex manni</i> Smith, 1946	SSP		1.25 *	1.25 *
<i>Strumigenys biolleyi</i> Forel, 1908	DP		1.25 *	
<i>Strumigenys cordovens</i> Mayr, 1887	DP		2.5 *	7.5 *
<i>Strumigenys delticus</i> Brown, 1957	DP	41.25 *	21.25	17.5
<i>Strumigenys denticulata</i> Mayr, 1887	DP	1.25 *		40
<i>Strumigenys dyseides</i> Bolton, 2000	DP	1.25 *		2.5 *
<i>Strumigenys eggersi</i> Emery, 1890	DP	55 *	56.25	37.5 *
<i>Strumigenys elongata</i> Roger, 1863	DP	35 *	43.75	40
<i>Strumigenys fridericimuelleri</i> Forel, 1886	DP	22.5 *	7.5 *	33.75 *
<i>Strumigenys lanuginosa</i> Wheeler, 1905	DP	5 *	11.25 *	3.75 *
<i>Strumigenys marginiventris</i> Santschi, 1931	DP	15 *	1.25	10
<i>Strumigenys subdentata</i> Mayr, 1887	DP		2.5	6.25 *
<i>Strumigenys zeteki</i> (Brown, 1959)	DP	2.5 *	40 *	3.75 *
<i>Trichomyrmex destructor</i> (Jerdon, 1851)	ASO	3.75	2.5	8.75 *
<i>Wasmannia auropunctata</i> (Roger, 1863)	SO	28.75 *	28.75	38.75 *
Ponerinae				
<i>Anochetus diegensis</i> Forel, 1912	MLO		21.25	21.25 *
<i>Anochetus inermis</i> André, 1889	MLO	11.25 *	1.25	
<i>Hypoponera opacior</i> (Forel, 1893)	SLP	28.75 *	56.25 *	33.75 *

Table A1. Cont.

Taxa	Functional Groups	N1	N2	N3
<i>Hypoponera trigona</i> (Mayr, 1887)	SLP	3.75 *	25 *	26.25 *
<i>Leptogenys ritae</i> Forel, 1899	SSI	6.25 *	3.75	1.25 *
<i>Mayaponera constricta</i> (Mayr, 1884)	MLO		6.25	22.5 *
<i>Mayaponera arhuaca</i> (Forel, 1901)	MLO	1.25 *	6.25 *	18.75 *
<i>Neoponera apicalis</i> (Latreille, 1802)	LEP		1.25	3.75 *
<i>Neoponera carinulata</i> (Roger, 1861)	LAP		2.5 *	1.25 *
<i>Neoponera verenae</i> (Forel, 1922)	LEP		1.25	
<i>Odontomachus bauri</i> Emery, 1892	LEP	2.5	5	10
<i>Odontomachus brunneus</i> (Patton, 1894)	LEP		2.5	2.5 *
<i>Odontomachus laticeps</i> Roger, 1861	LEP	2.5 *	3.75 *	
<i>Odontomachus opaciventris</i> Forel, 1899	LEP	15 *	26.25 *	6.25 *
<i>Pachycondyla harpax</i> (Fabricius, 1804)	LEP	26.25	36.25	18.75
<i>Pachycondyla impressa</i> (Roger, 1861)	LEP	2.5	3.75	5 *
<i>Rasopone pluviselva</i> Longino & Branstetter, 2020	SLP			1.25 **
<i>Thaumatomyrmex atrox</i> Weber, 1939	SSM	2.5	1.25	
<i>Thaumatomyrmex zeteki</i> Smith, 1944	SSM		6.25 *	18.75 *
Proceratiinae				
<i>Discothyrea humilis</i> Weber, 1939	SSP	3.75 *		1.25 *
<i>Discothyrea neotropica</i> Bruch, 1919	SSP	1.25 *	13.75	
<i>Discothyrea</i> sp. 4	SSP		1.25	
<i>Discothyrea testacea</i> Roger, 1863	SSP		1.25 *	12.5 *
<i>Probolomyrmex boliviensis</i> Mann, 1923	SSP		2.5 *	1.25 *
<i>Proceratium catio</i> de Andrade, 2003	SSP			1.25 *
Pseudomyrmecinae				
<i>Pseudomyrmex boopis</i> (Roger, 1863)	LAP		1.25	2.5
<i>Pseudomyrmex mordax</i> (Warming, 1894)	LAP		1.25	
<i>Pseudomyrmex simplex</i> (Smith, 1877)	LAP	1.25 *	1.25	1.25
<i>Pseudomyrmex</i> sp. 4	LAP	1.25		

References

- Pizano, C.; Cabrera, M.; García, H. Bosque seco tropical en Colombia; generalidades y contexto. In *El Bosque Seco Tropical en Colombia*; Pizano, C., García, H., Eds.; Instituto de Investigación de Recursos Biológicos Alexander von Humboldt (IAvH): Bogotá, Colombia, 2014; pp. 36–47.
- Ulloa-Delgado, G. *Aspectos Ecológicos del Bosque Seco Tropical en el Caribe Colombiano*; Tropenbos Internacional Colombia y Fondo Patrimonio Natural: Bogotá, Colombia, 2016; pp. 5–56.
- Janzen, D.H. Management of Habitat Fragments in a Tropical Dry Forest: Growth. *Ann. Mo. Bot. Gard.* **1988**, *75*, 105–116. [CrossRef]
- Zuluaga, L. Evaluación Estructural del Ecosistema Bosque Seco Tropical en el Municipio de El Carmen de Bolívar (Bolívar) y Determinación de sus Beneficios Ecosistémicos. Master's Thesis, Universidad de Manizales, Manizales, Colombia, 23 January 2016. Available online: <https://ridum.umanizales.edu.co/xmlui/handle/20.500.12746/2576> (accessed on 20 March 2024).
- Portillo-Quintero, C.A.; Sánchez-Azofeifa, G.A. Extent and conservation of tropical dry forests in the Americas. *Biol. Conserv.* **2010**, *143*, 144–155. [CrossRef]
- García, H.; González-M, R. *Bosque Seco Colombia: Biodiversidad y Gestión*; Instituto de Investigación de Recursos Biológicos Alexander von Humboldt: Bogotá, Colombia, 2019; p. 8.
- Franco-Maass, S.; Regil-Gracia, H.H.; Ordóñez-Díaz, J.A. Dinámica de perturbación-recuperación de las zonas forestales en el Parque Nacional Nevado de Toluca. *Madera y Bosques* **2006**, *12*, 17–28. [CrossRef]
- Martínez, N.J.; Cañas, L.M.; Rangel, J.L.; Barraza, J.; Montes, J.; Blanco, O.R. Fich. *Bol. Mus. Entomol. U. Valle* **2010**, *11*, 21–30.
- Agosti, D.; Alonso, L.E. The ALL Protocol: A standard protocol for the collection of ground-dwelling ants. In *Ants: Standard Methods for Measuring and Monitoring Biodiversity*; Agosti, D., Majer, J.D., Alonso, L.E., Schultz, T.R., Eds.; Smithsonian Institution Press: Washington, DC, USA, 2000; pp. 204–206.
- Folgarait, P.J. Ant biodiversity and its relationship to ecosystem functioning: A review. *Biod. Cons.* **1998**, *7*, 1221–1244. [CrossRef]
- Hölldobler, B.; Wilson, E.O. Ecology and Behavior of the primitive cryptobiotic ant *Prionopelta amabilis* (Hymenoptera: Formicidae). *Insectes Sociaux* **1986**, *33*, 45–58. [CrossRef]
- Andersen, A.N. Sampling communities of ground-foraging ants: Pitfall catches compared with quadrat counts in an Australian tropical savanna. *Aust. J. Ecol.* **1991**, *16*, 273–279. [CrossRef]

13. Underwood, E.C.; Fisher, B.L. The role of ants in conservation monitoring: If, when, and how. *Biol. Conserv.* **2006**, *132*, 166–182. [[CrossRef](#)]
14. Kaspari, M.; Majer, J.D. Using ants to monitor environmental change. In *Ants: Standard Methods for Measuring Y Monitoring Biodiversity*; Agosti, J., Majer, J.D., Alonso, L., Schultz, T.R., Eds.; Smithsonian Institution: Washington, DC, USA, 2000; pp. 89–98.
15. Schultz, T.R.; McGlynn, T.P. The interactions of ants with others organisms. In *Ants: Standard Methods for Measuring y Monitoring Biodiversity*; Agosti, J., Majer, J.D., Alonso, L., Schultz, T.R., Eds.; Smithsonian Institution: Washington, DC, USA, 2000; pp. 35–44.
16. Moreno, C. *La Biodiversidad en un Mundo Cambiante: Fundamentos Teóricos y Metodológicos para su Estudio*; Universidad Autónoma del Estado de Hidalgo y Libermex: Ciudad de México, México, 2019.
17. Pey, B.; Nahmani, J.; Auclerc, A.; Capowiez, Y.; Cluzeau, D.; Cortet, J.; Decaëns, J.; Deharveng, L.; Dubs, F.; Joimel, S.; et al. Current use of and future needs for soil invertebrate functional traits in community ecology. *Basic Appl. Ecol.* **2014**, *15*, 194–206. [[CrossRef](#)]
18. Wong, M.K.L.; Guénard, B.; Lewis, O.T. Trait-based ecology of terrestrial arthropods. *Biol. Rev.* **2019**, *94*, 999–1022. [[CrossRef](#)]
19. Cadotte, M.W.; Carscadden, K.; Mirotchnick, N. Beyond species: Functional diversity and the maintenance of ecological processes and services. *J. Appl. Ecol.* **2011**, *48*, 1079–1087. [[CrossRef](#)]
20. Díaz, S.; Cabido, M. Vive la différence: Plant functional diversity matters to ecosystem processes. *Trends Ecol. Evol.* **2001**, *16*, 646–655. [[CrossRef](#)]
21. Lohbeck, M.; Poorter, L.; Paz, H.; Pla, L.; van Breugel, M.; Martínez-Ramos, M.; Bongers, F. Functional diversity changes during tropical forest succession. *Perspect. Plant Ecol. Evol. Syst.* **2012**, *14*, 89–96. [[CrossRef](#)]
22. Yates, M.L.; Andrew, N.R.; Binns, M.; Gibb, H. Morphological traits: Predictable responses to macrohabitats across a 300 km scale. *PeerJ* **2014**, *2*, 1–20. [[CrossRef](#)]
23. Rocha, F.; Bogar, A.; Ibañez Weemaels, A.; Jiaxin, H.; Liang, M.; Park, C.; Lee, R.H.; Khan, S.A.; Han, S.; NG, Y.L.; et al. Two sides of the same coin? Ants are ecosystem engineers and providers of ecosystem services. *Myrmecol. News.* **2024**, *34*, 129–157.
24. Elizalde, L.; Arbetman, M.; Arnan, X.; Eggleton, P.; Leal, I.R.; Lescano, M.N.; Saez, A.; Werenkraut, V.; Pirk, G.I. The ecosystem services provided by social insects: Traits, management tools and knowledge gaps. *Biol. Rev.* **2020**, *95*, 1418–1441. [[CrossRef](#)] [[PubMed](#)]
25. Bihn, J.H.; Gebauer, G.; Brandl, R. Loss of functional diversity of ant assemblages in secondary tropical forests. *Ecology* **2010**, *91*, 782–792. [[CrossRef](#)]
26. Silva, R.; Brandão, C. Morphological patterns and community organization in leaf-litter ant assemblages. *Ecol. Monogr.* **2010**, *80*, 107–124. [[CrossRef](#)]
27. Brandão, C.R.; Silva, R.R.; Delabie, J.H. Neotropical Ants (Hymenoptera) Functional Groups: Nutritional and Applied Implications. In *Insect Bioecology and Nutrition for Integrated Pest Management*; Panizzi, A., Parra, J.R., Eds.; CRC Press: Boca Raton, FL, USA, 2012; pp. 231–254. [[CrossRef](#)]
28. Violle, C.; Navas, M.L.; Vile, D.; Kazakou, E.; Fortunel, C.; Hummel, I.; Garnier, E. Let the concept of trait be functional! *Oikos* **2007**, *116*, 882–892. [[CrossRef](#)]
29. Westoby, M.; Wright, I.J. Land-plant ecology on the basis of functional traits. *Trends Ecol. Evol.* **2006**, *21*, 261–268. [[CrossRef](#)]
30. Drager, K.I.; Rivera, M.D.; Gibson, J.C.; Ruzi, S.A.; Hanisch, P.E.; Achury, R.; Suarez, A.V. Testing the predictive value of functional traits in diverse ant communities. *Ecol. Evol.* **2023**, *13*, e10000. [[CrossRef](#)] [[PubMed](#)]
31. Wiescher, P.T.; Pearce-Duvet, J.M.; Feener, D.H. Assembling an ant community: Species functional traits reflect environmental filtering. *Oecologia* **2012**, *169*, 1063–1074. [[CrossRef](#)]
32. Weiser, M.D.; Kaspari, M. Ecological morphospace of New World ants. *Ecol. Entomol.* **2006**, *31*, 131–142. [[CrossRef](#)]
33. Kaspari, M.; Weiser, M.D. The size–grain hypothesis and interspecific scaling in ants. *Funct. Ecol.* **1999**, *13*, 530–538. [[CrossRef](#)]
34. Fowler, H.G.; Forti, L.C.; Brandão, C.R.; Delabie, J.H.; Vasconcelos, H.L. Ecología nutricional de formigas. In *Ecología Nutricional Insetos*; Oanizzi, A.R., Parra, J.R., Eds.; Manole: Brasil, Brazil, 1991; pp. 131–223.
35. Marques, T.; Espírito-Santo, M.M.; Schoereder, J.H. Ant Assemblage Structure in a Secondary Tropical Dry Forest: The Role of Ecological Succession and Seasonality. *Sociobiology* **2017**, *64*, 261–275. [[CrossRef](#)]
36. Silva, L.; Souza, M.; Solar, R.R.C.; Neves, F.S. Ant diversity in Brazilian tropical dry forests across multiple vegetation domains. *Environ. Res. Lett.* **2017**, *12*, 035002. [[CrossRef](#)]
37. de Menezes, A.S.; Schmidt, F.A. Mechanisms of species coexistence and functional diversity of ant assemblages in forest and pasture habitats in southwestern Brazilian Amazon. *Sociobiology* **2020**, *67*, 33–40. [[CrossRef](#)]
38. Armbrrecht, I.; Chacón, P. Rareza y Diversidad de Hormigas en Fragmentos de Bosque Seco Colombianos y sus Matrices. *Biotropica* **1999**, *31*, 646–653. [[CrossRef](#)]
39. Armbrrecht, I.; Tischer, I.; Chacón, P. Nested subsets and partition patterns in ant assemblages (Hymenoptera, Formicidae) of Colombian dry forest fragments. *Pan-Pac. Entomol.* **2001**, *77*, 196–209.
40. Arcila-Cardona, A.; Osorio, A.M.; Bermúdez, C.; Chacón de Ulloa, P. Diversidad de hormigas cazadoras asociadas a los elementos del paisaje del bosque seco. In *Sistemática, Biogeografía y Conservación de las Hormigas Cazadoras de Colombia*; Jiménez, E., Fernández, F., Arias, T., Lozano-Zambrano, F., Eds.; Instituto de Investigación de Recursos Biológicos Alexander von Humboldt: Bogotá, Colombia, 2008; pp. 531–552.
41. Chacón de Ulloa, P.; Osorio-García, A.M.; Achury, R.; Bermúdez-Rivas, C. Hormigas (Hymenoptera: Formicidae) del Bosque seco Tropical (Bs-T) de la cuenca alta del río Cauca, Colombia. *Biota Colomb.* **2012**, *13*, 165–181.

42. Gallego-Roper, M.C.; Salguero, B. Ensamblaje de hormigas del bosque seco tropical, jardín botánico de Cali. *Colomb. For.* **2015**, *18*, 139–150. [\[CrossRef\]](#)
43. Dix, O.; Martínez, J.; Fernández, C. Contribución al conocimiento de la mirmecofauna en el municipio de San Antero, cordoba, Colombia. *Rev. Colomb. Entomol.* **2005**, *31*, 97–103. [\[CrossRef\]](#)
44. Domínguez, Y.; Fontalvo, L.; Guitérrez, L.C. Composición y distribución espacio-temporal de las hormigas cazadoras (Formicidae: Grupos Poneroides y Ectatomminoides) en tres fragmentos de bosque seco tropical del departamento del Atlántico, Colombia. In *Sistemática, Biogeografía y Conservación de las Hormigas Cazadoras de Colombia*; Jiménez, E., Fernández, F., Arias, T.M., Lozano-Zambrano, F., Eds.; Instituto de Investigación de Recursos Biológicos Alexander von Humboldt: Bogotá, Colombia, 2007; pp. 497–512.
45. Fontalvo-Rodríguez, L.; Solís-Medina, C. Ensamblaje de hormigas (Hymenoptera: Formicidae) en fragmentos de bosque seco en el complejo carbonífero el Cerrejón (La Guajira, Colombia). *Rev. Introp.* **2009**, *4*, 5–15.
46. Simanca-Fontalvo, R.; Fajardo-Herrera, R.J. Fauna de hormigas (Hymenoptera: Formicidae) en dos remanentes de Bosque Seco Tropical (Bs-T) en Corrales de San Luis, Atlántico, Colombia. *Bol. Mus. Entomol. U. Valle* **2013**, *14*, 1.
47. Domínguez-Haydar, Y.; Armbrrecht, I. Response of Ants and Their Seed Removal in Rehabilitation Areas and Forests at El Cerrejón Coal Mine in Colombia. *Restor. Ecol.* **2011**, *19*, 178–184. [\[CrossRef\]](#)
48. Ramos Ortega, L.; Guerrero, R. Spatial Turnover and Functional Redundancy in the Ants of Urban Fragments of Tropical Dry Forest. *Diversity* **2023**, *15*, 880. [\[CrossRef\]](#)
49. Aguilera-Díaz, M.M. Montes de María: Una subregión de economía campesina y empresarial. *Documentos de Trabajo Sobre Economía Regional y Urbana* **2013**, 195.
50. Hernández, C.; Hurtado, G.; Ortiz, Q.; Walschulger, C. Unidades biogeográficas de Colombia. In *La Diversidad Biológica de Iberoamérica*; Halffter, G., Ed.; Acta Zoológica Mexicana: Ciudad de México, Mexico, 1992; Volume 1, pp. 100–115.
51. Guerrero, R.; Delsinne, T.; Dekoninck, W. Métodos de recolección y curadurías. In *Hormigas de Colombia*; Fernández, F., Guerrero, R., Delsinne, T., Eds.; Universidad Nacional de Colombia: Bogotá, Colombia, 2019; pp. 319–369.
52. Jost, L. Entropy and diversity. *Oikos* **2006**, *113*, 363–375. [\[CrossRef\]](#)
53. Hill, M.O. Diversity and Evenness: A Unifying Notation and Its Consequences. *Ecology* **1973**, *54*, 427–432. [\[CrossRef\]](#)
54. Hsieh, T.C.; Ma, K.H.; Chao, A. iNEXT: An R package for rarefaction and extrapolation of species diversity (Hill numbers). *Methods Ecol. Evol.* **2016**, *7*, 1451–1456. [\[CrossRef\]](#)
55. Chao, A.; Jost, L. Coverage-based rarefaction and extrapolation: Standardizing samples by completeness rather than size. *Ecology* **2012**, *93*, 2533–2547. [\[CrossRef\]](#) [\[PubMed\]](#)
56. Cultid, C.; Escobar, F. Pautas para la estimación y comparación estadística de la diversidad biológica (^qD). In *La Biodiversidad en un Mundo Cambiante: Fundamentos Teóricos y Metodológicos Para su Estudio*; Moreno, C., Ed.; Universidad Autónoma del Estado de Hidalgo: Pachuca de Soto, Mexico, 2019; pp. 175–202.
57. Schenker, N.; Gentleman, J.F. On judging the significance of differences by examining the overlap between confidence intervals. *Am. Stat.* **2001**, *55*, 182–186. [\[CrossRef\]](#)
58. Chao, A.; Gotelli, N.; Hsieh, T.C.; Sander, E.; Ma, K.H.; Colwell, R.; Ellison, A. Rarefaction and extrapolation with Hill numbers: A framework for sampling and estimation in species diversity studies. *Ecol. Monogr.* **2014**, *84*, 45–67. [\[CrossRef\]](#)
59. Baselga, A.; Orme, C.D. Betapart: An R package for the study of beta diversity. *Methods Ecol. Evol.* **2012**, *3*, 808–812. [\[CrossRef\]](#)
60. Baselga, A. Partitioning the turnover and nestedness components of beta diversity. *Glob. Ecol. Biogeogr.* **2010**, *19*, 134–143. [\[CrossRef\]](#)
61. Branstetter, M.G.; Ješovnik, A.; Sosa-Calvo, J.; Lloyd, M.W.; Faircloth, B.C.; Brady, S.G.; Schultz, T.R. Dry habitats were crucibles of domestication in the evolution of agriculture in ants. *Proc. R. Soc. Biol.* **2017**, *284*, 20170095. [\[CrossRef\]](#)
62. Koch, E.B.A.; dos Santos, J.R.M.; Nascimento, I.C.; Delabie, J.H.C. Comparative evaluation of taxonomic and functional diversities of leaf-litter ants of the Brazilian Atlantic Forest. *Turk. J. Zool.* **2019**, *43*, 437–546. [\[CrossRef\]](#)
63. Moretti, M.; Dias, A.T.; de Bello, F.; Altermatt, F.; Chown, S.L.; Azcárate, F.M.; Bell, J.R.; Fournier, B.; Hedde, M.; Hortal, J.; et al. Handbook of protocols for standardized measurement of terrestrial invertebrate functional traits. *Funct. Ecol.* **2017**, *31*, 558–567. [\[CrossRef\]](#)
64. Martello, F.; De Bello, F.; De Castro Morini, M.S.; Silva, R.R.; De Souza-Campana, D.R.; Ribeiro, M.C.; Carmona, C.P. Homogenization and impoverishment of taxonomic and functional diversity of ants in Eucalyptus plantations. *Sci. Rep.* **2018**, *8*, 3266. [\[CrossRef\]](#)
65. Parr, C.L.; Dunn, R.R.; Sanders, N.J.; Weiser, M.D.; Photakis, M.; Bishop, T.R.; Fitzpatrick, M.C.; Arnan, X.; Baccaro, F.; Brandão, C.R.F.; et al. GlobalAnts: A new database on the geography of ant traits (Hymenoptera: Formicidae). *Insect Conserv. Diver.* **2017**, *10*, 5–20. [\[CrossRef\]](#)
66. Sarty, M.; Abbott, K.L.; Lester, P.J. Habitat complexity facilitates coexistence in a tropical ant community. *Oecologia* **2006**, *149*, 465–473. [\[CrossRef\]](#) [\[PubMed\]](#)
67. Kaspari, M. Body size and microclimate use in Neotropical granivorous ants. *Oecologia* **1993**, *96*, 500–507. [\[CrossRef\]](#)
68. Gibb, H.; Cunningham, S.A. Restoration of trophic structure in an assemblage of omnivores, Considering a revegetation chronosequence. *J. Appl. Ecol.* **2013**, *50*, 449–458. [\[CrossRef\]](#)
69. Gibb, H.; Parr, C.L. Does Structural Complexity Determine the Morphology of Assemblages? An Experimental Test on Three Continents. *PLoS ONE* **2013**, *8*, e64005. [\[CrossRef\]](#)

70. Feener, D.H.; Lighton, J.R.; Bartholomew, G.A. Curvilinear Allometry, Energetics and Foraging Ecology: A Comparison of Leaf-Cutting Ants and Army Ants. *Funct. Ecol.* **1988**, *2*, 509–520. [\[CrossRef\]](#)
71. Gibb, H.; Stoklosa, J.; Warton, D.I.; Brown, A.M.; Andrew, N.R.; Cunningham, S.A. Does morphology predict trophic position and habitat use of ant species and assemblages? *Oecologia* **2015**, *177*, 519–531. [\[CrossRef\]](#) [\[PubMed\]](#)
72. Garnier, E.; Cortez, J.; Billès, G.; Navas, M.L.; Roumet, C.; Debussche, M.; Laurent, G.; Blanchard, A.; Aubry, D.; Bellmann, A.; et al. Plant functional markers capture ecosystem properties during secondary succession. *Ecology* **2004**, *85*, 2630–2637. [\[CrossRef\]](#)
73. Almeida, R.P.; Teresa, F.B.; Camarota, F.; Izzo, T.J.; Silva, R.R.; Andrade-Silva, J.; de Arruda, F.V. The role of morphological traits in predicting the functional ecology of arboreal and ground ants in the Cerrado–Amazon transition. *Oecologia* **2023**, *201*, 199–212. [\[CrossRef\]](#) [\[PubMed\]](#)
74. Vasconcelos, H.L.; Neves, K.C.; Vieira, J.; Carvalho, R.L. Land use intensification has extensive effects on the functional and phylogenetic diversity of neotropical ant communities. *Biodivers. Conserv.* **2024**, *33*, 2487–2502. [\[CrossRef\]](#)
75. Casanoves, F.; Pla, L.; Di Rienzo, J.A. *Valoración y Análisis de la Diversidad Funcional y su Relación con los Servicios Ecosistémicos*; Serie Técnica Informe Técnico; Centro Agronómico Tropical de Investigación y Enseñanza (CATIE): Turrialba, Costa Rica, 2011; pp. 1–85.
76. Oksanen, J.; Kindt, R.; Legendre, P.; O'Hara, B.; Stevens, M.H.; Oksanen, M.J.; Suggests, M. The vegan package. *Community Ecol. Package* **2007**, *10*, 631–637.
77. Fernández, F.; Guerrero, R.J.; Sánchez-Restrepo, A.F. Systematics and diversity of Neotropical ants. *Rev. Col. Entomol.* **2021**, *47*, e11082. [\[CrossRef\]](#)
78. Fernández, F.; Guerrero, R.; Delsinne, T. *Hormigas de Colombia*; Universidad Nacional de Colombia: Bogotá, Colombia, 2019; pp. 387–1113. [\[CrossRef\]](#)
79. Fernández, F.; Guerrero, R.; Delsinne, T. (Eds.) Filogenia y sistemática de las hormigas neotropicales. In *Hormigas de Colombia*; Universidad Nacional de Colombia: Bogotá, Colombia, 2019; pp. 57–89.
80. García, E.I.; Tocora, M.C.; Fiorentino, G.; Escárraga, M.E.; Fernández, F.; Guerrero, R.J. Nuevos registros de hormigas (Hymenoptera: Formicidae) para Colombia. *Biota Neotrop.* **2020**, *20*, e20201088.
81. Ward, P.S.; Brady, S.G.; Fisher, B.L.; Schultz, T.R. The evolution of myrmicine ants: Phylogeny and biogeography of a hyperdiverse ant clade (Hymenoptera: Formicidae). *Syst. Entomol.* **2015**, *40*, 61–81. [\[CrossRef\]](#)
82. Pennington, T.; Lavin, M.; Oliveira-Filho, A. Woody plant diversity, evolution, and ecology in the tropics: Perspectives from seasonally dry tropical forests. *Annu. Rev. Ecol. Evol. Syst.* **2009**, *40*, 437–457. [\[CrossRef\]](#)
83. Corro, E.J.; Ahuatzin, D.A.; Jaimes, A.A.; Favila, M.E.; Ribeiro, M.C.; López-Acosta, J.C.; Dáttilo, W. Forest cover and landscape heterogeneity shape ant–plant co-occurrence networks in human-dominated tropical rainforests. *Landsc. Ecol.* **2019**, *34*, 93–104. [\[CrossRef\]](#)
84. Hethcoat, M.G.; King, B.J.; Fernández, F.; Ortiz-Sepúlveda, C.M.; Achiardi, F.C.; Edwards, F.A.; Medina, C.; Gilroy, J.J.; Haugaasen, T.; Edwards, D.P. The impact of secondary forest regeneration on ground-dwelling ant communities in the Tropical Andes. *Oecologia* **2019**, *191*, 475–482. [\[CrossRef\]](#) [\[PubMed\]](#)
85. Kuchenbecker, J.; Cuevas-Reyes, P.; Fagundes, M. Community structure of ants (Hymenoptera: Formicidae) in an open habitat: The importance of environmental heterogeneity and interspecific interactions. *Rev. Mex. Biodiver.* **2022**, *93*, e933900. [\[CrossRef\]](#)
86. Siqueira, E.L.; Silva, R.R. Taxonomic and morphological diversity of the ground-dwelling ant fauna in Eastern Amazonian grasslands. *Acta Oecologica* **2021**, *110*, 103693. [\[CrossRef\]](#)
87. Delsinne, T.; Sonet, G.; Nagy, Z.T.; Wauters, N.; Jacquemin, J.; Leponce, M. High species turnover of the ant genus *Solenopsis* (Hymenoptera: Formicidae) along an altitudinal gradient in the Ecuadorian Andes, indicated by a combined DNA sequencing and morphological approach. *Invert. Syst.* **2012**, *26*, 457–469. [\[CrossRef\]](#)
88. Longino, J.T. A revision of the ant genus *Octostruma* Forel 1912 (Hymenoptera, Formicidae). *Zootaxa* **2013**, *3699*, 1–61. [\[CrossRef\]](#)
89. Calle, Z.; Henao-Gallego, N.; Giraldo, C.; Armbricht, I. A comparison of vegetation and ground-dwelling ants in abandoned and restored gullies and landslide surfaces in the Western Colombian Andes. *Restor. Ecol.* **2013**, *21*, 729–735. [\[CrossRef\]](#)
90. Chantásig-Vaca, C.; Huerta Lwanga, E.; Rojas Fernández, P.; Ponce-Mendoza, A.; Mendoza Vega, J.; Morón Ríos, A.; Van der Waal, H.; Dzib-Castillo, B. Efecto del uso de suelo en las hormigas (Formicidae: Hymenoptera) de Tikinmul, Campeche, México. *Acta Zool. Mex.* **2011**, *27*, 4441–4461. [\[CrossRef\]](#)
91. Silva, P.; Bieber, A.; Corrêa, M.; Leal, I. Do leaf-litter attributes affect the richness of leaf-litter ants? *Neotrop. Entomol.* **2011**, *40*, 542–547. [\[PubMed\]](#)
92. Mesa-s, L.; Santamaría, M.; García, H.; Aguilar-Cano, J. Catálogo de biodiversidad de la región caribe. In *Serie Planeación Ambiental para la Conservación de la Biodiversidad en Áreas Operativas de Ecopetrol*, 3rd ed.; Instituto de Investigación de Recursos Biológicos Alexander von Humboldt—Ecopetrol S.A: Bogotá, Colombia, 2016.
93. De La Vega, G.J.; Schilman, P.E. La importancia de la fisiología en la distribución geográfica de los insectos. *Rev. Soc. Entomol. Arg.* **2015**, *74*, 101–108.
94. Giweta, M. Role of litter production and its decomposition, and factors affecting the processes in a tropical forest ecosystem: A review. *J. Ecol. Environ.* **2020**, *44*, 11. [\[CrossRef\]](#)
95. Pennington, R.; Prado, D.E.; Pendry, C.A. Neotropical seasonally dry forests and Quaternary vegetation changes. *J. Biogeogr.* **2000**, *27*, 261–273. [\[CrossRef\]](#)
96. Dirzo, R. *Seasonally Dry Tropical Forests: Ecology and Conservation*; Island Press: Washington, DC, USA, 2011; pp. 1–408.

97. Angulo, E.; Boulay, R.; Ruano, F.; Tinaut, A.; Cerdá, X. Anthropogenic impacts in protected areas: Assessing the efficiency of conservation efforts using Mediterranean ant communities. *PeerJ* **2016**, *4*, e2773. [[CrossRef](#)]
98. Fernández, F.; y Sendoya, S. Synonymic list of neotropical ants (Hymenoptera: Formicidae). *Biot. Col.* **2004**, *5*, 3–105.
99. Kaspari, M.; O'Donnell, S. High rates of army ant raids in the Neotropics and implications for ant colony and community structure. *Evol. Ecol. Res.* **2003**, *5*, 933–939.
100. Abadía, J.C.; Bermúdez, C.; Lozano-Zambrano, F.H.; Chacón, P. Hormigas cazadoras en un paisaje subandino de Colombia: Riqueza, composición y especies indicadoras. *Rev. Col. Ent.* **2010**, *36*, 127–134. [[CrossRef](#)]
101. Chacón de Ulloa, P.; Armbrrecht, I.; Lozano-Zambrano, F.; Jiménez, E.; Fernández, F.; Arias, T.M. Aspectos de la ecología de hormigas cazadoras en bosques secos colombianos. In *Sistemática, Biogeografía y Conservación de las Hormigas Cazadoras de Colombia*; Jiménez, E., Fernández, F., Arias, T., Lozano-Zambrano, F., Eds.; Instituto de Investigación de Recursos Biológicos Alexander von Humboldt: Bogotá, Colombia, 2008; pp. 513–531.
102. Jiménez, B.; De la Rosa, N.; Naranjo, D.; Bustillo, R.; Parra, J.; Villalba, I.; Buelvas, C.; Pacheco, E.; Arrieta, J.; Castro, C.; et al. *Plan de Manejo del Santuario de Flora y Fauna Los Colorados 2018–2023*; Parques Nacionales Naturales de Colombia: Bogotá, Colombia, 2018.
103. Bustos, J.; Ulloa-Chacón, P. Mirmecofauna y perturbación en un bosque de niebla neotropical (Reserva Natural Hato Viejo, Valle del Cauca, Colombia). *Rev. Biol. Trop.* **1996**, *44*, 259–266.
104. Rivera, L.; Armbrrecht, I. Diversidad de tres gremios de hormigas en cafetales de sombra, de sol y bosques de Risaralda. *Rev. Col. Entomol.* **2005**, *31*, 89–96. [[CrossRef](#)]
105. Mason, N.W.; MacGillivray, K.; Steel, J.B.; Wilson, J.B. An index of functional diversity. *J. Veg. Sci.* **2003**, *14*, 571–578. [[CrossRef](#)]
106. Carmona, C.P.; Guerrero, I.; Morales, M.B.; Oñate, J.J.; Peco, B. Assessing vulnerability of functional diversity to species loss: A case study in Mediterranean agricultural systems. *Funct. Ecol.* **2017**, *31*, 427–435. [[CrossRef](#)]
107. Ricotta, C.; de Bello, F.; Moretti, M.; Caccianiga, M.; Cerabolini, B.E.; Pavoine, S. Measuring the functional redundancy of biological communities: A quantitative guide. *Methods Ecol. Evol.* **2016**, *7*, 1386–1395. [[CrossRef](#)]
108. Leps, J.; de Bello, F.; Lavorel, S.; Berman, S. Quantifying and interpreting functional diversity of natural communities: Practical considerations matter. *Preslia* **2006**, *78*, 481–501.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.

Copyright of Diversity (14242818) is the property of MDPI and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.